

SkeletonSR (BasicSR): Pseudocode

Overview. The BasicSR algorithm skeleton. The *policy* (in BLUE) is the evolvable configuration of the BasicSR skeleton, consisting of operators for calculating `LOSS_FUNCTION`, `SELECTION`, `MUTATION`, `CROSSOVER`, `ACCEPTANCE`, `SURVIVAL`, `UPDATE_POPULATION` and `UPDATE_STATE!`. The overall skeleton structure and constants (population sizes, number of iterations, etc.) originate from the PySR algorithm implementation.

```
1: function FITSKELETONSR( $X, y$ , config)
2:   policy_state  $\leftarrow$  init_state(config)
3:   populations  $\leftarrow$  15 populations of 33 small random trees
4:   UPDATE_STATE!(populations)
5:   for iteration = 1 ... 100 do
6:     for each population  $P$  do
7:       for 1 ... 380 do
8:         UPDATE_STATE!(populations)
9:         for 1 ... 3 do
10:          if rand  $\leq$  0.0259 then
11:             $a \leftarrow$  SELECTION( $P$ );  $b \leftarrow$  SELECTION( $P$ ) (distinct)
12:            children  $\leftarrow$  CROSSOVER( $a, b$ ); compute LOSS_FUNCTION(child) for each;
            keep those passing ACCEPTANCE(parent, child)
13:            if children non-empty then  $P \leftarrow$  SURVIVAL( $P$ , children)
14:          else
15:            parent  $\leftarrow$  SELECTION( $P$ )
16:            child  $\leftarrow$  MUTATION(parent); compute LOSS_FUNCTION(child)
17:            if ACCEPTANCE(parent, child) then  $P \leftarrow$  SURVIVAL( $P$ , [child])
18:          UPDATE_STATE!(populations)
19:        populations  $\leftarrow$  UPDATE_POPULATION(populations)
20:   return policy_state.archive
```